

What's up with 'verbal' morphology in BCS agent nominals?

Maša Bešlin, University of Maryland
mbeslin@umd.edu



FDSL 17, Masaryk University
November 21, 2024

Introduction

SLIDE FOR QR CODE

Introduction

MAIN CLAIM:

‘Verbal’ morphology in Bosnian/Croatian/Serbian agentive nouns is not verbal.

Introduction

- I'll be looking at a sample of BCS agentive nominals, which contain what is traditionally analyzed as verbal morphology

- | | | | | | | |
|-----|----|---|----|---|----|--|
| (1) | a. | pozn-av-á-telj
know-AV-TH-N
'expert' | b. | prouč-av-á-telj
study-AV-TH-N
'researcher' | c. | reš-av-á-telj
solve-AV-TH-N
'solver' |
| (2) | a. | predsed-av-a-áč
chair-AV-TH-N
'chair' | b. | pred-av-a-áč
lecture-AV-TH-N
'lecturer' | c. | ugnjjet-av-a-áč
oppress-AV-TH-N
'oppressor' |
| (3) | a. | prod-av-a-ác
sell-AV-TH-N
'seller' | b. | dar-o-d-av-a-ác
gift-L-give-AV-TH-N
'giftgiver' | c. | posl-o-d-av-a-ác
job-L-give-AV-TH-N
'employer' |

- A noun like *proučavatelj* is often segmented as *pro-uč-a-va-telj* 'LP-learn-V-SI-N', because of the similar verbs *proučavati* 'be researching', *proučiti*, and *učiti*

Introduction

- I'll be looking at a sample of BCS agentive nominals, which contain what is traditionally analyzed as verbal morphology

(1)	a. pozn-av-á-telj know-AV-TH-N 'expert'	b. prouč-av-á-telj study-AV-TH-N 'researcher'	c. reš-av-á-telj solve-AV-TH-N 'solver'
(2)	a. predsed-av-a-áč chair-AV-TH-N 'chair'	b. pred-av-a-áč lecture-AV-TH-N 'lecturer'	c. ugnjet-av-a-áč oppress-AV-TH-N 'oppressor'
(3)	a. prod-av-a-ác sell-AV-TH-N 'seller'	b. dar-o-d-av-a-ác gift-L-give-AV-TH-N 'giftgiver'	c. posl-o-d-av-a-ác job-L-give-AV-TH-N 'employer'

- A noun like *proučavatelj* is often segmented as *pro-uč-a-va-telj* 'LP-learn-V-SI-N', because of the similar verbs *proučavati* 'be researching', *proučiti*, and *učiti*
- ★ We'll see reasons to doubt that these nouns have verbal structure

Introduction

- (4) a. pozn-av-á-telj b. prouč-av-á-telj c. řeš-av-á-telj
 know-AV-TH-N study-AV-TH-N solve-AV-TH-N
 ‘expert’ ‘researcher’ ‘solver’
- (5) a. predsed-av-a-áč b. pred-av-a-áč c. ugnjet-av-a-áč
 chair-AV-TH-N lecture-AV-TH-N oppress-AV-TH-N
 ‘chair’ ‘lecturer’ ‘oppressor’
- (6) a. prod-av-a-ác b. dar-o-d-av-a-ác c. posl-o-d-av-a-ác
 sell-AV-TH-N gift-L-give-AV-TH-N job-L-give-AV-TH-N
 ‘seller’ ‘giftgiver’ ‘employer’

- Notice: (I) different *n*-allomorphs, and (II) the accent of *-áč* and *-ác* surfaces

Introduction

- (4) a. pozn-av-á-telj b. prouč-av-á-telj c. řeš-av-á-telj
 know-AV-TH-N study-AV-TH-N solve-AV-TH-N
 ‘expert’ ‘researcher’ ‘solver’
- (5) a. predsed-av-a-áč b. pred-av-a-áč c. ugnjet-av-a-áč
 chair-AV-TH-N lecture-AV-TH-N oppress-AV-TH-N
 ‘chair’ ‘lecturer’ ‘oppressor’
- (6) a. prod-av-a-ác b. dar-o-d-av-a-ác c. posl-o-d-av-a-ác
 sell-AV-TH-N gift-L-give-AV-TH-N job-L-give-AV-TH-N
 ‘seller’ ‘giftgiver’ ‘employer’

- Notice: (I) different *n*-allomorphs, and (II) the accent of -áč and -ác surfaces
- I’ll show BCS root-conditioned allomorphy and accent placement are limited to the **first spellout domain, including only one categorizing morpheme**

Introduction

- | | | | | | | |
|-----|----|---|----|---|----|--|
| (4) | a. | pozn-av-á-telj
know-AV-TH-N
'expert' | b. | prouč-av-á-telj
study-AV-TH-N
'researcher' | c. | reš-av-á-telj
solve-AV-TH-N
'solver' |
| (5) | a. | predsed-av-a-áč
chair-AV-TH-N
'chair' | b. | pred-av-a-áč
lecture-AV-TH-N
'lecturer' | c. | ugnj-av-a-áč
oppress-AV-TH-N
'oppressor' |
| (6) | a. | prod-av-a-ác
sell-AV-TH-N
'seller' | b. | dar-o-d-av-a-ác
gift-L-give-AV-TH-N
'giftgiver' | c. | posl-o-d-av-a-ác
job-L-give-AV-TH-N
'employer' |

- Notice: (I) different *n*-allomorphs, and (II) the accent of -áč and -ác surfaces
- I'll show BCS root-conditioned allomorphy and accent placement are limited to the **first spellout domain, including only one categorizing morpheme**
- ★ Then, the 'verbal' morphology in these agent nominals may not be verbal after all

Roadmap

- §1 Some background on Distributed Morphology (DM), cyclic domains, the role of categorizers, and allomorphy

Roadmap

- §1 Some background on Distributed Morphology (DM), cyclic domains, the role of categorizers, and allomorphy
- §2 Data from root-derived* vs. deadjectival agent nominals (Bešlin forthcoming)

Roadmap

- §1 Some background on Distributed Morphology (DM), cyclic domains, the role of categorizers, and allomorphy
- §2 Data from root-derived* vs. deadjectival agent nominals (Bešlin forthcoming)
 - **(I) Root-conditioned allomorphy** and **(II) accent placement** determined in the first spellout domain, centered around the first-merged categorizer
- Second-merged categorizer can't 'see' the root and can't realize its accent
- This follows from a DM conception of cyclic domains

Roadmap

- §1 Some background on Distributed Morphology (DM), cyclic domains, the role of categorizers, and allomorphy
- §2 Data from root-derived* vs. deadjectival agent nominals (Bešlin forthcoming)
 - **(I) Root-conditioned allomorphy** and **(II) accent placement** determined in the first spellout domain, centered around the first-merged categorizer
- Second-merged categorizer can't 'see' the root and can't realize its accent
- This follows from a DM conception of cyclic domains
- §3 Back to 'deverbal' agentive nouns...

Roadmap

- §1 Some background on Distributed Morphology (DM), cyclic domains, the role of categorizers, and allomorphy
- §2 Data from root-derived* vs. deadjectival agent nominals (Bešlin forthcoming)
 - **(I) Root-conditioned allomorphy** and **(II) accent placement** determined in the first spellout domain, centered around the first-merged categorizer
- Second-merged categorizer can't 'see' the root and can't realize its accent
- This follows from a DM conception of cyclic domains
- §3 Back to 'deverbal' agentive nouns...
 - A general note about verbal structure in agentive nouns

Roadmap

- §1 Some background on Distributed Morphology (DM), cyclic domains, the role of categorizers, and allomorphy
- §2 Data from root-derived* vs. deadjectival agent nominals (Bešlin forthcoming)
 - **(I) Root-conditioned allomorphy** and **(II) accent placement** determined in the first spellout domain, centered around the first-merged categorizer
- Second-merged categorizer can't 'see' the root and can't realize its accent
- This follows from a DM conception of cyclic domains
- §3 Back to 'deverbal' agentive nouns...
 - A general note about verbal structure in agentive nouns
 - They pattern in (I) and (II) with root-derived nouns (one categorizer)
 - An alternative analysis for 'verbal' morphemes ($\neq$ verbal extended projection)

Roadmap

- §1 Some background on Distributed Morphology (DM), cyclic domains, the role of categorizers, and allomorphy
- §2 Data from root-derived* vs. deadjectival agent nominals (Bešlin forthcoming)
 - **(I) Root-conditioned allomorphy** and **(II) accent placement** determined in the first spellout domain, centered around the first-merged categorizer
- Second-merged categorizer can't 'see' the root and can't realize its accent
- This follows from a DM conception of cyclic domains
- §3 Back to 'deverbal' agentive nouns...
 - A general note about verbal structure in agentive nouns
 - They pattern in (I) and (II) with root-derived nouns (one categorizer)
 - An alternative analysis for 'verbal' morphemes ($\neq$ verbal extended projection)
- The morpheme *av* as a root (also Quaglia et al. 2022);
- Theme vowels as morphemes that attach to (certain) roots more generally;
- 'Lexical prefixes' observed in contexts in which a deverbal analysis is dubious

Roadmap

- §1 Some background on Distributed Morphology (DM), cyclic domains, the role of categorizers, and allomorphy
- §2 Data from root-derived* vs. deadjectival agent nominals (Bešlin forthcoming)
- **(I) Root-conditioned allomorphy** and **(II) accent placement** determined in the first spellout domain, centered around the first-merged categorizer
- Second-merged categorizer can't 'see' the root and can't realize its accent
- This follows from a DM conception of cyclic domains
- §3 Back to 'deverbal' agentive nouns...
- A general note about verbal structure in agentive nouns
 - They pattern in (I) and (II) with root-derived nouns (one categorizer)
 - An alternative analysis for 'verbal' morphemes ($\neq$ verbal extended projection)
- The morpheme *av* as a root (also Quaglia et al. 2022);
- Theme vowels as morphemes that attach to (certain) roots more generally;
- 'Lexical prefixes' observed in contexts in which a deverbal analysis is dubious
- §4 Conclusions

Theoretical background & assumptions

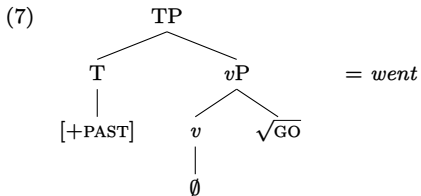
- DM is a piece-based, realizational approach to morphology

Theoretical background & assumptions

- DM is a piece-based, realizational approach to morphology
- Words are built up syntactically out of (discrete) abstract morphemes which receive form (and meaning) at the interfaces

Theoretical background & assumptions

- DM is a piece-based, realizational approach to morphology
- Words are built up syntactically out of (discrete) abstract morphemes which receive form (and meaning) at the interfaces
- Morphemes: roots and functional heads (including categorizers)
- The form (and meaning) of a morpheme may be contextually determined, (7)
- Allomorphs are in competition with each other ('Elsewhere principle')



Theoretical background & assumptions

- Transfer to the interfaces happens cyclically, at certain points of the derivation
- Categorizers (v , n , a) are the relevant cyclic heads

(8) Schematization of cyclic domains (Embick 2014):

a. Cyclic y merged in $[y [X [Y [x \sqrt{\text{ROOT}} \dots]]]]$

b. Cyclic domain centered on $x = [X [Y [x \sqrt{\text{ROOT}}]]]$ sent to interfaces

Theoretical background & assumptions

- Transfer to the interfaces happens cyclically, at certain points of the derivation
- Categorizers (v , n , a) are the relevant cyclic heads

(8) Schematization of cyclic domains (Embick 2014):

a. Cyclic y merged in $[y [X [Y [x \sqrt{\text{ROOT}} \dots]]]]$

b. Cyclic domain centered on $x = [X [Y [x \sqrt{\text{ROOT}}]]]$ sent to interfaces

→ Intended outcome: The root is accessible to the first cyclic head x and any intervening non-cyclic heads (X , Y) (think *go-went*)

→ $\sqrt{\text{ROOT}}$ and y cannot interact for the purposes of allomorph selection because they are in separate spell-out domains

Theoretical background & assumptions

- Cyclic spellout is thought to explain many patterns of (im)possible morphophonological interactions, including (im)possible allomorphy

Theoretical background & assumptions

- Cyclic spellout is thought to explain many patterns of (im)possible morphophonological interactions, including (im)possible allomorphy
- The root-conditioned *n*-allomorphy in (9) compared to (10) is thought to arise due to a structural difference: root-derived versus deverbal nouns

(9) marri-**age**, grow-**th**, remov-**al**, free-**dom**, divers-**ity**, strateg-**y**, ...

(10) marry-**ing**, grow-**ing**, remov-**ing**, free-**ing**, divers-ify-**ing**, strateg-iz-**ing**, ...

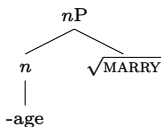
Theoretical background & assumptions

- Cyclic spellout is thought to explain many patterns of (im)possible morphophonological interactions, including (im)possible allomorphy
- The root-conditioned *n*-allomorphy in (9) compared to (10) is thought to arise due to a structural difference: root-derived versus deverbal nouns

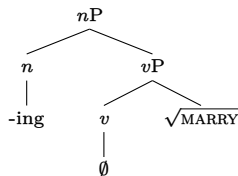
(9) marri-**age**, grow-**th**, remov-**al**, free-**dom**, divers-**ity**, strateg-**y**, ...

(10) marry-**ing**, grow-**ing**, remov-**ing**, free-**ing**, divers-ify-**ing**, strateg-iz-**ing**, ...

(11) a.



b.

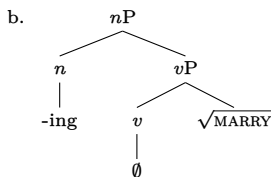
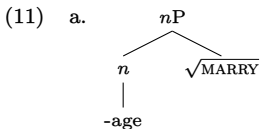


Theoretical background & assumptions

- Cyclic spellout is thought to explain many patterns of (im)possible morphophonological interactions, including (im)possible allomorphy
- The root-conditioned *n*-allomorphy in (9) compared to (10) is thought to arise due to a structural difference: root-derived versus deverbal nouns

(9) marri-**age**, grow-**th**, remov-**al**, free-**dom**, divers-**ity**, strateg-**y**, ...

(10) marry-**ing**, grow-**ing**, remov-**ing**, free-**ing**, divers-ify-**ing**, strateg-iz-**ing**, ...



→ Categorization has the same effect on allomorphy and accent placement in BCS

Root-nominals vs. deadjectival nominals

Root-nominals vs. deadjectival nominals

- I'll demonstrate that BCS *a* is a cyclic head—it imposes a locality boundary for morphophonological processes (Bešlin forthcoming):

Root-nominals vs. deadjectival nominals

- I'll demonstrate that BCS *a* is a cyclic head—it imposes a locality boundary for morphophonological processes (Bešlin forthcoming):
 - Root-conditioned allomorphy
 - Accent placement

Root-nominals vs. deadjectival nominals

- I'll demonstrate that BCS *a* is a cyclic head—it imposes a locality boundary for morphophonological processes (Bešlin forthcoming):
 - Root-conditioned allomorphy
 - Accent placement
- I'll contrast the behavior of deadjectival and root-derived (agentive) nouns

Root-nominals vs. deadjectival nominals

- I'll demonstrate that BCS *a* is a cyclic head—it imposes a locality boundary for morphophonological processes (Bešlin forthcoming):
 - Root-conditioned allomorphy
 - Accent placement
- I'll contrast the behavior of deadjectival and root-derived (agentive) nouns
- In §3, we'll see that 'deverbal' agentive nouns pattern with root-derived nouns

Root-nominals vs. deadjectival nominals: ALLOMORPHY

- Looking again at agent nominals, the broadly agentive *n*-suffixes in BCS are at least *-ar*, *-aš*, *-er*, *-(a)c*, *-ač*, *-ic(a)*, *-ik*, and *-džij(a)*
- Root-derived nouns (**ROOT-*n***) may take any of the *n*-allomorphs on offer; the choice of *n* is determined by the root ('lexically-conditioned allomorphy')

(12) a. kormil-**ar** 'helmsman'

b. batin-**aš** 'beater'

c. poz-**er** 'poser'

d. pis-**ac** 'writer'

e. voz-**ač** 'driver'

f. izdaj-**ica** 'traitor'

g. proza-**ik** 'prose writer'

h. bureg-**džija** 'börek maker'

Root-nominals vs. deadjectival nominals: ALLOMORPHY

- Looking again at agent nominals, the broadly agentive *n*-suffixes in BCS are at least *-ar*, *-aš*, *-er*, *-(a)c*, *-ač*, *-ic(a)*, *-ik*, and *-džij(a)*
- Root-derived nouns (**ROOT-*n***) may take any of the *n*-allomorphs on offer; the choice of *n* is determined by the root ('lexically-conditioned allomorphy')

(12) a. kormil-**ar** 'helmsman'

b. batin-**aš** 'beater'

c. poz-**er** 'poser'

d. pis-**ac** 'writer'

e. voz-**ač** 'driver'

f. izdaj-**ica** 'traitor'

g. proza-**ik** 'prose writer'

h. bureg-**džija** 'börek maker'

- NB:** I clearly do not subscribe to the view that all agentive nouns contain verbal structure, even if they seem to correspond to the external argument of a verb

Root-nominals vs. deadjectival nominals: ALLOMORPHY

- A root may determine the choice of n only if n is the first-merged categorizer

Root-nominals vs. deadjectival nominals: ALLOMORPHY

- A root may determine the choice of n only if n is the first-merged categorizer
- If a intervenes between the root and n (**ROOT- a - n**), the root can no longer determine the form of n (13)-(16)

Root-nominals vs. deadjectival nominals: ALLOMORPHY

- A root may determine the choice of *n* only if *n* is the first-merged categorizer
- If *a* intervenes between the root and *n* (**ROOT-*a*-*n***), the root can no longer determine the form of *n* (13)-(16)

- (13)
- | | | | |
|----|--------------------------------|----|-------------------------------------|
| a. | prlj- <i>av-ac</i> ‘dirty one’ | d. | peg- <i>av-ac</i> ‘freckled one’ |
| b. | mrš- <i>av-ac</i> ‘skinny one’ | e. | prg- <i>av-ac</i> ‘grumpy one’ |
| c. | mut- <i>av-ac</i> ‘mute one’ | f. | hvalis- <i>av-ac</i> ‘boastful one’ |

Root-nominals vs. deadjectival nominals: ALLOMORPHY

- A root may determine the choice of *n* only if *n* is the first-merged categorizer
- If *a* intervenes between the root and *n* (**ROOT-*a*-*n***), the root can no longer determine the form of *n* (13)-(16)

- (13) a. *prlj-av-ac* ‘dirty one’
b. *mrš-av-ac* ‘skinny one’
c. *mut-av-ac* ‘mute one’

- d. *peg-av-ac* ‘freckled one’
e. *prg-av-ac* ‘grumpy one’
f. *hvalis-av-ac* ‘boastful one’

- (14) a. *plaš-ljiv-ac* ‘scared one’
b. *smrd-ljiv-ac* ‘stinky one’
c. *grab-ljiv-ac* ‘predatory one’

- d. *povod-ljiv-ac* ‘gullible one’
e. *var-ljiv-ac* ‘cheating one’
f. *vaš-ljiv-ac* ‘lousy one’

Root-nominals vs. deadjectival nominals: ALLOMORPHY

- (15) a. hajduk-*ov-ac* ‘H. supporter’
b. dinam-*ov-ac* ‘D. supporter’
c. isus-*ov-ac* ‘Jesuit’
d. maček-*ov-ac* ‘Maček follower’
e. nobel-*ov-ac* ‘Nobel winner’
f. oskar-*ov-ac* ‘Oscar winner’
- (16) a. smrt-*n-ik* ‘mortal one’
b. put-*n-ik* ‘traveler’
c. boles-*n-ik* ‘sick one’
d. bestid-*n-ik* ‘shameless one’
e. duž-*n-ik* ‘debtor’
f. gubit-*n-ik* ‘loser’

- Only *a* can now influence the form of *n*, which is uniform regardless of the root in question (either due to *a*-conditioned allomorphy or *elsewhere*)

Root-nominals vs. deadjectival nominals: ALLOMORPHY

- The locality effect is best observed when the same root can produce both a root-nominal and a deadjectival nominal (cf. **gubit-n-aš*, **gubit-ik*)
- Same root, same meaning, different nominalizer due to the presence of *a*

(17) a. *gubit-aš*
lose-N
'loser'

b. *gubit-n-ik*
lose-A-N
'loser'

Root-nominals vs. deadjectival nominals: ALLOMORPHY

- The locality effect is best observed when the same root can produce both a root-nominal and a deadjectival nominal (cf. **gubit-n-aš*, **gubit-ik*)
- Same root, same meaning, different nominalizer due to the presence of *a*

(17) a. *gubit-aš*
lose-N
'loser'

b. *gubit-n-ik*
lose-A-N
'loser'

- Even though the root $\sqrt{gubit}$ clearly picks out the nominalizer *-aš*, it can no longer do so if an adjektivizer intervenes between the two

Root-nominals vs. deadjectival nominals: ALLOMORPHY

- The locality effect is best observed when the same root can produce both a root-nominal and a deadjectival nominal (cf. **gubit-n-aš*, **gubit-ik*)
- Same root, same meaning, different nominalizer due to the presence of *a*

(17) a. *gubit-aš*
lose-N
'loser'

b. *gubit-n-ik*
lose-A-N
'loser'

- Even though the root $\sqrt{gubit}$ clearly picks out the nominalizer *-aš*, it can no longer do so if an adjectivizer intervenes between the two
- This can be accounted for if *a* and *n* are cyclic heads

Root-nominals vs. deadjectival nominals: ALLOMORPHY

- The locality effect is best observed when the same root can produce both a root-nominal and a deadjectival nominal (cf. **gubit-n-aš*, **gubit-ik*)
- Same root, same meaning, different nominalizer due to the presence of *a*

(17) a. *gubit-aš*
lose-N
'loser'

b. *gubit-n-ik*
lose-A-N
'loser'

- Even though the root $\sqrt{gubit}$ clearly picks out the nominalizer *-aš*, it can no longer do so if an adjektivizer intervenes between the two
 - This can be accounted for if *a* and *n* are cyclic heads
- In a **ROOT-*a*-*n*** configuration, the root is spelled out when *n* is merged, hence the root (*qua* morpheme) can no longer be identified when *n* undergoes VI

Root-nominals vs. deadjectival nominals: ACCENT

Root-nominals vs. deadjectival nominals: ACCENT

- **Exponents** of BCS morphemes are idiosyncratically (un)marked for accent*

Root-nominals vs. deadjectival nominals: ACCENT

- **Exponents** of BCS morphemes are idiosyncratically (un)marked for accent*
- Bešlin (forthcoming): Pitch-prominence in BCS is realized on the **structurally highest accent-marked element in the first spellout domain**

Root-nominals vs. deadjectival nominals: ACCENT

- **Exponents** of BCS morphemes are idiosyncratically (un)marked for accent*
- Bešlin (forthcoming): Pitch-prominence in BCS is realized on the **structurally highest accent-marked element in the first spellout domain**
- The nominalizer $-(\acute{a})c$ is underlyingly accent-marked, but only realizes that accent if it is in **ROOT-*n***, not in e.g., **ROOT-*a-n***

(18) a. $\text{pis} \rightarrow \text{pis-}\acute{a}c$
 $\sqrt{\text{write}}$ 'writer'

b. $\text{al}\acute{z}\acute{i}r \rightarrow \text{al}\acute{z}\acute{i}r-}\acute{a}c$
 $\sqrt{\text{algeria}}$ 'Algerian(N)'

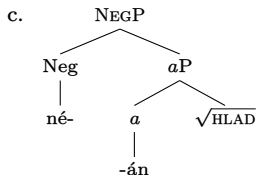
(19) a. $\text{p}\acute{f}lj\text{-}av \rightarrow \text{p}\acute{f}lj\text{-}av\text{-}ac$
 'dirty' 'dirty one'

b. $\text{smrd}\text{-}lj\acute{i}v \rightarrow \text{smrd}\text{-}lj\acute{i}v\text{-}ac$
 'stinky' 'stinky one'

Root-nominals vs. deadjectival nominals: ACCENT

- If the extended projection of the first categorizer contains non-cyclic heads (e.g., DEG, NEG, DIM) and their exponents are accented, the accent surfaces on them

- (20) a. hlad-án
cold-A
'cold'
- b. né-hlad-an
NEG-cold-A
'non-cold'

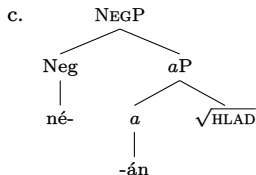


Root-nominals vs. deadjectival nominals: ACCENT

- If the extended projection of the first categorizer contains non-cyclic heads (e.g., DEG, NEG, DIM) and their exponents are accented, the accent surfaces on them

(20) a. hlad-án
cold-A
'cold'

b. né-hlad-an
NEG-cold-A
'non-cold'



- Accent placement is determined within the first spellout domain, as in (8)/(21)

(21) Schematization of cyclic domains (Embick 2014):

a. Cyclic y merged in [y [X [Y [x $\sqrt{\text{ROOT}}$...]]]

b. Cyclic domain centered on x = [X [Y [x $\sqrt{\text{ROOT}}$]]] sent to interfaces

Root-nominals vs. deadjectival nominals: ACCENT

- Root-root compounds behave as expected; the accent placement is still ‘frozen’ in the spellout domain of the first **categorizer**

(22) a. dub-o-rez-ác
deep-L-cut-N
‘woodcarver’

b. pad-o-bran-ác
fall-L-defend-N
‘parachuter’

c. led-o-lom-ác
ice-L-break-N
‘ice-breaker’

Root-nominals vs. deadjectival nominals

INTERIM SUMMARY I:

Root-conditioned allomorphy and accent placement in BCS are limited to the first spellout domain, including *one* categorizer

‘Deverbal’ nouns

§1 Some background on Distributed Morphology (DM), cyclic domains, the role of categorizers, and allomorphy

§2 Data from root-derived* vs. deadjectival agent nominals (Bešlin forthcoming)

- (I) **Root-conditioned allomorphy** and (II) **accent placement** determined in the first spellout domain, centered around the first-merged categorizer

→ Second-merged categorizer can’t ‘see’ the root and can’t realize its accent

→ This follows from a DM conception of cyclic domains

§3 Back to ‘deverbal’ agentive nouns...

- A general note about verbal structure in agentive nouns
- They pattern in (I) and (II) with root-derived nouns (one categorizer)
- An alternative analysis for ‘verbal’ morphemes ($\neq$ verbal extended projection)

→ The morpheme *av* as a root (also Quaglia et al. 2022);

→ Theme vowels as morphemes that attach to (certain) roots more generally;

→ ‘Lexical prefixes’ observed in contexts in which a deverbal analysis is dubious

§4 Conclusions

Agentive noun $\neq$ verbal structure

- Originally, eventive (episodic) interpretation = complement structure = verbal syntax (23a), but cf. (23b-c)

- (23) a. a frequent consumer *(of tobacco)
 b. a frequent visitor
 c. a frequent subject *(of Monet's paintings)

- Then, Alexiadou & Schäfer 2010 propose that both episodic and dispositional agent nominals have an articulated verbal structure, including *v*, Voice, and Asp
- Why no accusative case on complements or adverbial modification?
- Completely divorced from the syntax, accounting for the fact that the *-er* nominals denote the (external) argument of the corresponding verb
- But we know agent entailments $\neq$ Voice, cf. *hastily* in (24)

- (24) The rock rolled down the hill quickly/#hastily.

Agentive noun $\neq$ verbal structure

- Moreover, if *painter* needs Voice, does *thief* too?
- Event entailments also don't implicate the presence of *v*; cf. (25a)/(25b-c), (26)

- (25) a. a beautiful dancer
 b. a beautiful violinist
 c. an elegant midfielder

- (26) a. a just ruler
 b. a just king

- At the broadest level, entailments $\neq$ the presence of hidden structure

- (27) a. an illegitimate blond child

‘Deverbal’ nouns: ALLOMORPHY

‘Deverbal’ nouns: ALLOMORPHY

- As we saw in the beginning, there are different *n*-allomorphs in ‘deverbal’ nouns

- (28) a. **pozn-av-a-**telj**** b. **prouč-av-a-**telj**** c. **reš-av-a-**telj****
know-AV-TH-N study-AV-TH-N solve-AV-TH-N
‘expert’ ‘researcher’ ‘solver’
- (29) a. **predsed-av-a-**ač**** b. **pred-av-a-**ač**** c. **ugnjjet-av-a-**ač****
chair-AV-TH-N lecture-AV-TH-N oppress-AV-TH-N
‘chair’ ‘lecturer’ ‘oppressor’
- (30) a. **prod-av-a-**ac**** b. **dar-o-d-av-a-**ac**** c. **posl-o-d-av-a-**ac****
sell-AV-TH-N gift-L-give-AV-TH-N job-L-give-AV-TH-N
‘seller’ ‘giftgiver’ ‘employer’

‘Deverbal’ nouns: ALLOMORPHY

- As we saw in the beginning, there are different *n*-allomorphs in ‘deverbal’ nouns

- (28) a. **pozn-av-a-telj** b. **prouč-av-a-telj** c. **reš-av-a-telj**
 know-AV-TH-N study-AV-TH-N solve-AV-TH-N
 ‘expert’ ‘researcher’ ‘solver’
- (29) a. **predsed-av-a-ač** b. **pred-av-a-ač** c. **ugnjjet-av-a-ač**
 chair-AV-TH-N lecture-AV-TH-N oppress-AV-TH-N
 ‘chair’ ‘lecturer’ ‘oppressor’
- (30) a. **prod-av-a-ac** b. **dar-o-d-av-a-ac** c. **posl-o-d-av-a-ac**
 sell-AV-TH-N gift-L-give-AV-TH-N job-L-give-AV-TH-N
 ‘seller’ ‘giftgiver’ ‘employer’

→ The material in intervening between the root and *n* is the same

‘Deverbal’ nouns: ALLOMORPHY

- As we saw in the beginning, there are different *n*-allomorphs in ‘deverbal’ nouns

- (28) a. **pozn-av-a-telj** b. **prouč-av-a-telj** c. **reš-av-a-telj**
 know-AV-TH-N study-AV-TH-N solve-AV-TH-N
 ‘expert’ ‘researcher’ ‘solver’
- (29) a. **predsed-av-a-ač** b. **pred-av-a-ač** c. **ugnjjet-av-a-ač**
 chair-AV-TH-N lecture-AV-TH-N oppress-AV-TH-N
 ‘chair’ ‘lecturer’ ‘oppressor’
- (30) a. **prod-av-a-ac** b. **dar-o-d-av-a-ac** c. **posl-o-d-av-a-ac**
 sell-AV-TH-N gift-L-give-AV-TH-N job-L-give-AV-TH-N
 ‘seller’ ‘giftgiver’ ‘employer’

→ The material in intervening between the root and *n* is the same

→ No syntactic difference correlates with different *ns* (e.g., argument structure)

‘Deverbal’ nouns: ALLOMORPHY

- As we saw in the beginning, there are different *n*-allomorphs in ‘deverbal’ nouns

- (28) a. **pozn-av-a-telj** b. **prouč-av-a-telj** c. **reš-av-a-telj**
 know-AV-TH-N study-AV-TH-N solve-AV-TH-N
 ‘expert’ ‘researcher’ ‘solver’
- (29) a. **predsed-av-a-ač** b. **pred-av-a-ač** c. **ugnjjet-av-a-ač**
 chair-AV-TH-N lecture-AV-TH-N oppress-AV-TH-N
 ‘chair’ ‘lecturer’ ‘oppressor’
- (30) a. **prod-av-a-ac** b. **dar-o-d-av-a-ac** c. **posl-o-d-av-a-ac**
 sell-AV-TH-N gift-L-give-AV-TH-N job-L-give-AV-TH-N
 ‘seller’ ‘giftgiver’ ‘employer’

- The material in intervening between the root and *n* is the same
- No syntactic difference correlates with different *ns* (e.g., argument structure)
- No semantic or phonological factors that condition the allomorphy

‘Deverbal’ nouns: ALLOMORPHY

- As we saw in the beginning, there are different *n*-allomorphs in ‘deverbal’ nouns

- (28) a. **pozn-av-a-telj** b. **prouč-av-a-telj** c. **reš-av-a-telj**
 know-AV-TH-N study-AV-TH-N solve-AV-TH-N
 ‘expert’ ‘researcher’ ‘solver’
- (29) a. **predsed-av-a-ač** b. **pred-av-a-ač** c. **ugnjjet-av-a-ač**
 chair-AV-TH-N lecture-AV-TH-N oppress-AV-TH-N
 ‘chair’ ‘lecturer’ ‘oppressor’
- (30) a. **prod-av-a-ac** b. **dar-o-d-av-a-ac** c. **posl-o-d-av-a-ac**
 sell-AV-TH-N gift-L-give-AV-TH-N job-L-give-AV-TH-N
 ‘seller’ ‘giftgiver’ ‘employer’

- The material in intervening between the root and *n* is the same
- No syntactic difference correlates with different *ns* (e.g., argument structure)
- No semantic or phonological factors that condition the allomorphy
- ★ Allomorphy of *n* in (28)-(30) is lexically conditioned by the root

‘Deverbal’ nouns: ACCENT

‘Deverbal’ nouns: ACCENT

- As we also saw in the beginning, accent in our ‘deverbal’ nouns can surface on *n*-exponents that underlyingly have it (*-áč* and *-ác*):

- (31) a. pozn-av-á-telj b. prouč-av-á-telj c. řeš-av-á-telj
 know-AV-TH-N study-AV-TH-N solve-AV-TH-N
 ‘expert’ ‘researcher’ ‘solver’
- (32) a. predsed-av-a-áč b. pred-av-a-áč c. ugnjet-av-a-áč
 chair-AV-TH-N lecture-AV-TH-N oppress-AV-TH-N
 ‘chair’ ‘lecturer’ ‘oppressor’
- (33) a. prod-av-a-ác b. dar-o-d-av-a-ác c. posl-o-d-av-a-ác
 sell-AV-TH-N gift-L-give-AV-TH-N job-L-give-AV-TH-N
 ‘seller’ ‘giftgiver’ ‘employer’

‘Deverbal’ nouns: ACCENT

- As we also saw in the beginning, accent in our ‘deverbal’ nouns can surface on *n*-exponents that underlyingly have it (*-áč* and *-ác*):

- (31) a. pozn-av-á-telj b. prouč-av-á-telj c. řeš-av-á-telj
 know-AV-TH-N study-AV-TH-N solve-AV-TH-N
 ‘expert’ ‘researcher’ ‘solver’
- (32) a. predsed-av-a-áč b. pred-av-a-áč c. ugnjet-av-a-áč
 chair-AV-TH-N lecture-AV-TH-N oppress-AV-TH-N
 ‘chair’ ‘lecturer’ ‘oppressor’
- (33) a. prod-av-a-ác b. dar-o-d-av-a-ác c. posl-o-d-av-a-ác
 sell-AV-TH-N gift-L-give-AV-TH-N job-L-give-AV-TH-N
 ‘seller’ ‘giftgiver’ ‘employer’

- Recall, accent can only surface in the first spellout domain (one categorizer)

‘Deverbal’ nouns

INTERIM SUMMARY II:

Allomorphy and accent placement patterns suggest that the *n* in BCS ‘deverbal’ agentive nouns is the first-merged categorizer.

‘Deverbal’ nouns

INTERIM SUMMARY II:

Allomorphy and accent placement patterns suggest that the *n* in BCS ‘deverbal’ agentive nouns is the first-merged categorizer.

COROLLARY:

‘Verbal’ morphology inside these agentive nouns is not verbal.

What of the ‘verbal’ morphology, then? THE STATUS OF AV

What of the ‘verbal’ morphology, then? THE STATUS OF AV

- The morphemes exponed by *-av* and *-iv* appear in so-called secondary imperfective verbs and signal a shift in aspect (34)-(35)

(34) a. prouč-i-ti
study-TH-INF
‘research’

b. prouč-av-a-ti
study-AV-TH-INF
‘be researching’

(35) a. zatašk-a-ti
coverup-TH-INF
‘cover up’

b. zatašk-iv-a-ti
coverup-IV-TH-INF
‘be covering up’

What of the ‘verbal’ morphology, then? THE STATUS OF AV

- The morphemes expounded by *-av* and *-iv* appear in so-called secondary imperfective verbs and signal a shift in aspect (34)-(35)

(34) a. prouč-i-ti
study-TH-INF
‘research’

b. prouč-av-a-ti
study-AV-TH-INF
‘be researching’

(35) a. zatašk-a-ti
coverup-TH-INF
‘cover up’

b. zatašk-iv-a-ti
coverup-IV-TH-INF
‘be covering up’

- They also appear in *some* agent nominals; same meaning in (36a-b) vs. (36c-d)

(36) a. prouč-av-a-telj
study-AV-TH-N
‘researcher’

c. uruč-i-telj
serve-TH-N
‘process server’

b. zatašk-iv-a-ač
coverup-IV-TH-N
‘cover up agent’

d. istovar-a-ač
unload-TH-N
‘unloader’

What of the ‘verbal’ morphology, then? THE STATUS OF AV

- They can appear in the context of so-called verbs of creation
(cf. Kratzer 2000, Embick 2004)

(37) 3D štampač je pokvaren pa je maketa izašla
 3D printer is broken so is model came_out
 iz-u-ništ-av-a-n-a / is-pre-sav-ij-a-n-a.
 SP-LP-destroy-AV-TH-PTCP-F.SG SP-LP-bend-IJ-TH-PTCP-F.SG
 ‘The 3D printer is broken so the model came out destroyed/crumpled.’

What of the ‘verbal’ morphology, then? THE STATUS OF AV

- They can appear in the context of so-called verbs of creation
(cf. Kratzer 2000, Embick 2004)

(37) 3D štampač je pokvaren pa je maketa izašla
 3D printer is broken so is model came_out
 iz-u-ništ-av-a-n-a / is-pre-sav-ij-a-n-a.
 SP-LP-destroy-AV-TH-PTCP-F.SG SP-LP-bend-IJ-TH-PTCP-F.SG
 ‘The 3D printer is broken so the model came out destroyed/crumpled.’

- Quaglia et al. (2022): They also appear in the derivation of (seemingly) simple nouns and adjectives—they are **bound roots**

(38) a. ruk-av-∅
 arm-AV-N.M.SG.NOM
 ‘sleeve’
 b. bles-av-∅
 silly-AV-A.M.SG.NOM
 ‘silly’
 c. maz-iv-o
 daub-IV-N.NEUT.SG.NOM
 ‘grease’
 d. jez-iv-o
 shudder-IV-A.NEUT.SG.NOM
 ‘creepy’

What of the ‘verbal’ morphology, then? THE STATUS OF AV

STATUS OF AV

I tentatively conclude with Quaglia et al. (2022) that *av* is a root.

What of the ‘verbal’ morphology, then? THE STATUS OF TH

What of the ‘verbal’ morphology, then? THE STATUS OF TH

- Theme vowels (THs) do not seem to be a uniquely verbal phenomenon

What of the ‘verbal’ morphology, then? THE STATUS OF TH

- Theme vowels (THs) do not seem to be a uniquely verbal phenomenon
- Slavic nouns have also been claimed to have theme vowels
(Halle 1994, Bailyn & Nevins 2008, Halle & Nevins 2009, a.o.)

What of the ‘verbal’ morphology, then? THE STATUS OF TH

- Theme vowels (THs) do not seem to be a uniquely verbal phenomenon
- Slavic nouns have also been claimed to have theme vowels
(Halle 1994, Bailyn & Nevins 2008, Halle & Nevins 2009, a.o.)
- Slavic adjectives have been claimed to share the THs of nouns
(Halle & Matushansky 2006)

What of the ‘verbal’ morphology, then? THE STATUS OF TH

- Theme vowels (THs) do not seem to be a uniquely verbal phenomenon
- Slavic nouns have also been claimed to have theme vowels
(Halle 1994, Bailyn & Nevins 2008, Halle & Nevins 2009, a.o.)
- Slavic adjectives have been claimed to share the THs of nouns
(Halle & Matushansky 2006)

If all major word classes have THs, THs could then equally well be attributed to roots, as in (39) (with contextual allomorphy able to work in the familiar way)

- (39)
- a. ROOT-TH-*n*
 - b. ROOT-TH-*a*
 - c. ROOT-TH-*v*
 - d. ROOT-TH-ROOT

What of the ‘verbal’ morphology, then? THE STATUS OF TH

- Theme vowels (THs) do not seem to be a uniquely verbal phenomenon
- Slavic nouns have also been claimed to have theme vowels
(Halle 1994, Bailyn & Nevins 2008, Halle & Nevins 2009, a.o.)
- Slavic adjectives have been claimed to share the THs of nouns
(Halle & Matushansky 2006)

If all major word classes have THs, THs could then equally well be attributed to roots, as in (39) (with contextual allomorphy able to work in the familiar way)

- (39)
- a. ROOT-TH-*n*
 - b. ROOT-TH-*a*
 - c. ROOT-TH-*v*
 - d. ROOT-TH-ROOT

→ If this is correct, then the appearance of a TH does not necessarily indicate the presence of a verbal categorizing morpheme

What of the ‘verbal’ morphology, then? THE STATUS OF LPS

What of the ‘verbal’ morphology, then? THE STATUS OF LPS

- Is the decomposition always synchronic? Experimental work needed:

(40)	a.	d-a-ti	b.	pro -d-a-ti	c.	pro -d-a-av-ac
		give-TH-INF		LP?-give-TH-INF		LP?-give-TH-AV-N
		‘give’		‘sell’		‘seller’

What of the ‘verbal’ morphology, then? THE STATUS OF LPS

- Is the decomposition always synchronic? Experimental work needed:

(40)	a. d-a-ti	b. pro -d-a-ti	c. pro -d-a-av-ac
	give-TH-INF	LP?-give-TH-INF	LP?-give-TH-AV-N
	‘give’	‘sell’	‘seller’

- LPs appear in all sorts of words for which a deverbal analysis is dubious (41)

(41)	a. na -uč-i-ti	b. na -uk-a	c. na -uk-∅
	LP?-study-TH-INF	LP?-learn-N.NOM.SG	LP?-learn-N.NOM.SG
	‘learn/teach’	‘science’	‘lesson’

→ Also *pred-stava* ‘play’, *pre-preka* ‘barrier’, *o-stava* ‘pantry’, *iz-reka* ‘proverb’, etc.

Wrapping up

- I argued, based on data from ROOT-*n* vs. ROOT-*a-n*, that root-conditioned allomorphy and accent placement are limited to the first spellout domain

Wrapping up

- I argued, based on data from ROOT-*n* vs. ROOT-*a-n*, that root-conditioned allomorphy and accent placement are limited to the first spellout domain
- The first spellout domain may include multiple morphemes (roots, non-categorizing morphemes), but **only one categorizer**

Wrapping up

- I argued, based on data from ROOT-*n* vs. ROOT-*a-n*, that root-conditioned allomorphy and accent placement are limited to the first spellout domain
- The first spellout domain may include multiple morphemes (roots, non-categorizing morphemes), but **only one categorizer**
- I showed that ‘deverbal’ agentive nouns containing morphology analyzed as verbal behave for these morphophonological processes like root-derived nouns

Wrapping up

- I argued, based on data from ROOT-*n* vs. ROOT-*a-n*, that root-conditioned allomorphy and accent placement are limited to the first spellout domain
- The first spellout domain may include multiple morphemes (roots, non-categorizing morphemes), but **only one categorizer**
- I showed that ‘deverbal’ agentive nouns containing morphology analyzed as verbal behave for these morphophonological processes like root-derived nouns
- No syntactic evidence for verbal structure in agentive nouns; event/agent entailments do not provide evidence either

Wrapping up

- I argued, based on data from ROOT-*n* vs. ROOT-*a-n*, that root-conditioned allomorphy and accent placement are limited to the first spellout domain
- The first spellout domain may include multiple morphemes (roots, non-categorizing morphemes), but **only one categorizer**
- I showed that ‘deverbal’ agentive nouns containing morphology analyzed as verbal behave for these morphophonological processes like root-derived nouns
- No syntactic evidence for verbal structure in agentive nouns; event/agent entailments do not provide evidence either
- I argued that *av* should be analyzed as a root, and suggested that THs and LPS may not necessarily signal the presence of verbal structure either

- I'm grateful to Masha Polinsky, Dave Embick, Tanja Milićev, Norbert Hornstein, Alex Chabot, Bill Idsardi, Heather Newell, Tobias Scheer, Hannah Sande, Jim Wood, the audience at NELS 55, & the participants of Yale's Syntax Reading Group and UMD's S-lab for valuable discussion and feedback on various aspects of this work.

- I'm grateful to Masha Polinsky, Dave Embick, Tanja Milićev, Norbert Hornstein, Alex Chabot, Bill Idsardi, Heather Newell, Tobias Scheer, Hannah Sande, Jim Wood, the audience at NELS 55, & the participants of Yale's Syntax Reading Group and UMD's S-lab for valuable discussion and feedback on various aspects of this work.

Thank you!

References

- Alexiadou, Artemis & Florian Schäfer. 2010. On the syntax of episodic vs. dispositional -er nominals, 9–39. Berlin: Walter de Gruyter.
- Bailyn, John & Andrew Nevins. 2008. Russian genitive plurals are impostors. In A. Bachrach & A. Nevins (eds.), Inflectional identity, 237–270. Oxford: Oxford University Press.
- Bešlin, Maša. forthcoming. Lexical categories, (re)categorization, and locality in morphosyntax: University of Maryland dissertation.
- Embick, David. 2004. On the Structure of Resultative Participles in English. Linguistic Inquiry 35(3). 355–392.
- Embick, David. 2014. Phase cycles, ϕ -cycles, and phonological (in)activity. In S. Bendjaballah, M. Lahrouchi, N. Faust & N. Lampitelli (eds.), The Form of Structure, the Structure of Form, 270–286. Amsterdam: John Benjamins.
- Halle, Morris. 1994. The Russian declension. In J. Cole & C. Kisserberth (eds.), Perspectives in phonology, 29–60. Stanford, CA: CSLI Publications.
- Halle, Morris. 1997. On stress and accent in Indo-European. Language 73. 275–313.
- Halle, Morris & Ora Matushansky. 2006. The morphophonology of Russian adjectival inflection. Linguistic inquiry 37(3). 351–404.
- Halle, Morris & Andrew Nevins. 2009. Rule application in phonology. In E. Raimy & C. E. Cairns (eds.), Contemporary Views on Architecture and Representations in Phonology, 355–383. Cambridge, MA: MIT Press.
- Kratzer, Angelika. 2000. Building Statives. In L. J. Conathan, J. Good, D. Kavitskaya, A. Wulf & M. Yu (eds.), Proceedings of the 26th Annual Meeting of the Berkeley Linguistics Society, 385–399. University of California, Berkeley Linguistics Society.
- Quaglia, Stefano, Marko Simonović, Svitlana Antonyuk Yudina & Boban Arsenijević. 2022. Allomorphy, morphological operations and the order of slavic verb-prefixes. Journal of Slavic linguistics 30(3). 1–15.

Appendix A: Post-accenting elements

- There is a group of examples that form a systematic exception to the accent generalizations made here—so-called “post-accenting” elements (Halle 1997)
- Have an underlying accent, but realize it on the syllable following them
- Assuming the existence of such elements allows us to avoid having a list of pairs of suffixes that are segmentally identical and only differ in presence/absence of accent (uniform for two kinds of roots); root in (a)-(g) is post-accenting

(42)	a.	loz-á	‘grape-N.NOM.SG.F’	h.	dúnj-a	‘quince-N.NOM.SG.F’
	b.	loz-é	‘grape-N.GEN.SG.FEM’	i.	dúnj-e	‘quince-N.GEN.SG.F’
	c.	loz-í	‘grape-N.DAT.SG.F’	j.	dúnj-i	‘quince-N.DAT.SG.F’
	d.	loz-ú	‘grape-N.ACC.SG.F’	k.	dúnj-u	‘quince-N.ACC.SG.F’
	e.	loz-óm	‘grape-N.INST.SG.F’	l.	dúnj-om	‘quince-N.INST.SG.F’
	f.	loz-í.ca	‘grape-N.DIM-F’	m.	dúnj-ic-a	‘quince-N.DIM-F’
	g.	loz-óv	‘grape-A.POSS’	n.	dúnj-ev	‘quince-A.POSS’